# Chapter 3
# Cambodian Cassava and COVID-19: An Analysis of Production, Productivity, and Gender Impacts

**Kosal Nith and Yuki Kanayama**

**Abstract** Although the agriculture sector in Cambodia was initially considered less vulnerable to the COVID-19 pandemic, cassava farmers experienced severe economic hardship due to the increase in input prices, fluctuations in output prices, and challenges in infrastructure development. However, these aggregate impacts hide the disproportionate repercussions faced by cash crop farmers, who were more severely affected by public health restrictions and market disruptions. Using data from 301 cassava-farming households across five major cultivation provinces, this study examines the pandemic's impact on production, labor productivity, and labor participation in Cambodia. Although the price of fresh cassava slightly increased and the price of cassava chips remained stable, farmers lost income due to the increase in input prices and the fall in productivity. Between 2019 and 2021, average production costs rose by 9.3%, while income declined by 8.5%. Despite expanding cultivated land and increasing hired labor, overall productivity decreased. These findings underscore the vulnerability of Cambodia's cassava sector to external shocks and highlight the need for targeted policy interventions. The paper concludes with actionable recommendations to enhance value addition, strengthen domestic markets for cassava products, and support a resilient post-pandemic recovery.

**Keywords** Cassava · Agriculture · Production · Productivity · Gender differences · COVID-19

**JEL Codes** E23 · E24 · E61 · Q12

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## 3.1 Introduction

The COVID-19 pandemic represents the most significant upheaval to the global economy since the World War II era, causing a substantial and widespread decline in GDP. By mid-2020, Cambodia witnessed a surge in healthcare demand alongside severe economic setbacks. The Cambodian economy, heavily reliant on garment manufacturing, tourism, construction, and agriculture (Weggel, 2006; World Bank, 2023b),[1] was profoundly affected by disruptions in the global supply chain and the halt in international travel. This shock significantly altered Cambodia's economic trajectory, shifting from its average annual growth rate of 7% to a notable contraction of 3.1% in 2021 (IMF, 2022). The impact was uneven across economic sectors, with industries requiring high levels of social interaction—such as airlines, automobiles, hospitality, education, and retail—suffering the most (UNDP, 2021). The crisis has also necessitated a reorganization of production processes, with lasting implications for productivity, labor participation, and overall economic activity.

During the economic slowdown, while various sectors experienced reduced growth, agriculture proved relatively resilient to the pandemic. According to the World Bank (2023a), agriculture's share of Cambodia's GDP increased by 2 percentage points in 2020, reaching 22.7%. This growth continued into 2021 (22.8%) before slightly declining to 21.9% in 2022, still above the pre-pandemic level of 20.7% in 2019 (World Bank, 2023a). Overall, agriculture demonstrated resilience during challenging times for three key reasons. First, rural areas reported fewer cases of COVID-19 than urban centers, allowing farmers to continue activities such as cropping, livestock rearing, and aquaculture, largely uninterrupted (UNDP, 2020).[2] People could travel within their communities and visit their farms unless the lockdowns were enforced. Second, recognizing agriculture's relative stability, the government increased its support to the sector when other industries struggled (Barua, 2020). In 2020, the Agriculture and Rural Development Bank (ARDB) provided $50 million in financial subsidies to farmers (ARDB, 2020),[3] offering loans at a reduced annual interest rate of approximately 5% to help them sustain and improve their businesses (ARDB, 2021; 2022). Third, the agricultural workforce grew because thousands of migrant workers returned from Thailand and Cambodia's urban areas (Walkowiak, 2022). Having lost their jobs, many sought to assist their families in farming, further bolstering the sector.

Despite agriculture's overall expansion during the COVID-19 pandemic, questions remain about how different types of farmers coped with economic shocks.

[1] IMF (2022) estimated Cambodia's GDP growth rebounded to 3% in 2021, 5% in 2022 and 5.4% in 2023. The subsequent recovery has been sluggish compared to the pre-pandemic period.

[2] ADB (2021) and Chhim et al. (2023) provide a thorough analysis of government measures in response to curbing the pandemic within communities and supporting vulnerable groups.

[3] In October 2023, as part of the implementation of the Pentagonal Strategy, the government injected $100 million into the agricultural sector through the Ministry of Agriculture, Forestry and Fisheries to enhance farming production, expand markets, and maintain a balance between farming costs and prices.

In particular, those growing cash crops may have been disproportionately affected due to domestic and international mobility restrictions, that prevented buyers from collecting their produce. Hence, it is crucial to examine the heterogeneous effects of the pandemic and highlight the specific vulnerabilities faced by cash crop farmers.

This paper explores the impact of COVID-19 on cassava producers, focusing on production, productivity, and gender roles. The study draws on data from 301 cassava-producing households, collected through the Household Phone Survey conducted by the Centre for Development Economics and Trade (CDET) at the Cambodia Development Resource Institute (CDRI) in 2022. The primary objective is to assess how the pandemic influenced cassava farmers' gross income and production costs. Furthermore, we examine productivity and gender participation by analyzing cassava output and labor use. This study also evaluates government policy responses on cassava and offers recommendations to strengthen the sector.

Our contributions to the literature on COVID-19 and its socioeconomic impact on agriculture are twofold. First, while most studies analyze country-level or sector-wide impacts, we provide a micro-level analysis of household level effects in the cassava sector (Hing, 2023; Kansiime et al., 2021; Karamba et al., 2021; Rudin-Rush et al., 2022). Second, we contribute to the growing literature on agricultural policy analysis by reviewing existing cassava-related policies and discussing alternatives to support smallholder farmers (Alston & James, 2002; Ellis, 1992; Jackson et al., 2009).

This chapter is structured as follows: Sect. 3.2 provides background on cassava production in Cambodia. In Sect. 3.3, we present the research methodology, primary dataset, and sampling strategy. Section 3.4 presents the main findings, while Sect. 3.5 discusses these results in a broader context. In Sect. 3.6, we describe the current state of cassava policies and provide policy alternatives, and in the last section, we deliver a concluding remark. We also provide supplementary figures and information in the appendix.

## 3.2 Overview of Cassava Production and the Impact of COVID-19

### *3.2.1 Cassava and the Cambodian Economy*

Cassava plays a crucial role in the Cambodian economy. Its production expanded rapidly from 150,000 tons in 2000 to 8 million tons in 2011 (FAO, 2023). The sector relies heavily on external demand from Thailand, Vietnam, and China, making cassava an essential income source for rural Cambodian farmers.

Existing research from Kampong Thom and Ratanakiri demonstrates that small-scale farmers are particularly drawn to cassava production for several reasons (Beban & Gironde, 2023). First, the rapidly expanding value chain allows farmers to sell cassava more easily while acquiring skills and resources through trader-led training and in-kind credit (Mahanty & Milne, 2016). Second, improved road infrastructure and access to motorized tools have significantly improved market connectivity. Third, cassava is less labor-intensive yet more profitable compared to other common agricultural staple crops such as rice. It is also well-suited to Cambodia's climate and is relatively resilient to climate change. These advantages make cassava attractive to smallholders and a priority for policymakers seeking to support agricultural development.

Nonetheless, cassava production faces several challenges, particularly price volatility. Beban and Gironde (2023) report that the price of fresh cassava fluctuated significantly: 300 riel/kg in 2016, halving in 2017, rising back to 300 riel/kg in 2018, and dropping again in 2019. Such instability causes financial stress among farmers,[4] compounded by rising input costs and food prices. To deal with these challenges, some farmers allocated less financial capital and land to cassava while others stopped growing cassava altogether and pursued non-farm income opportunities (e.g., wage jobs or entrepreneurship). However, most smallholders lack the resources to diversify their crops and must continue growing cassava to sustain their livelihoods.

The COVID-19 pandemic exacerbated these challenges. Lockdowns and travel restrictions disrupted supply chains, preventing traders from accessing farms and resulting in income losses (World Farmers' Organization, 2020). The decline in agricultural income also led to underinvestment in healthcare and personal protective equipment, increasing their vulnerability to the virus. Furthermore, many young migrants who had worked in neighboring countries returned home due to job losses, reducing remittances, and placing additional financial pressure on rural households.

In response, the Cambodian government implemented several policies to support cassava producers. The Cambodia-China Free Trade Agreement, signed in October 2020 and enacted in January 2022, is expected to create new income opportunities by increasing cassava exports to China (Chheang, 2023). Additionally, in 2021, a national development policy was launched through a collaboration between various Cambodian ministries and the United Nations Development Programme (UNDP, 2021). Given these recent policy efforts, cassava is expected to remain a key crop for rural livelihoods.

### *3.2.2 COVID-19 and Agriculture*

Although agriculture was less affected by COVID-19 than other sectors, such as manufacturing and tourism, the pandemic nonetheless exposed vulnerabilities in farming communities around the world. Studies from Africa, Europe, and Asia show that farmers faced a range of disruptions including reduced access to markets, falling produce prices, labor shortages, rising input costs, and declining remittances. In

[4] In 2023, the price of cassava outputs increased significantly due to the high demand from China. The price of fresh cassava was in the range of 300–500 riel/kg while the price of dried cassava was between 800 and 900 riel/kg in early 2023 (Chantha, 2023).

Africa, smallholder farmers in Nigeria and across East Africa experienced income losses due to mobility restrictions, decreased off-farm employment, and disruptions to food supply chains (Bamidele & Amole, 2021; Hammond et al., 2022). Similarly, in Europe, farmers reported reduced consumer demand and difficulties accessing seasonal labor, particularly in high-value crops (Meredith et al., 2020). In Southeast Asia, Wulandari et al. (2022) and Sapbamrer et al. (2022) found that many farmers struggled with the affordability and availability of essential inputs like fertilizers and seeds, while lockdowns limited market access and mobility.

These constraints had cascading effects. Many farmers were unable to sell their produce, leading to declining incomes and heightened financial stress (O'Hara et al., 2021). At the same time, the rising cost of food and basic goods reduced household consumption and increased vulnerability to poverty and food insecurity (Nchanji & Kweyu, 2021). In several cases, these economic strains were associated with a deterioration in farmers' mental health, with higher levels of anxiety, depression, and stress reported among rural populations (David et al., 2020; Rose et al., 2023; Thompson et al., 2022).

Cambodia shares many of these characteristics, particularly in its reliance on smallholder farmers and cash crop production, such as cassava. As with other countries in the region, Cambodian farmers faced mobility restrictions and supply chain interruptions that reduced their ability to access markets and inputs. Many migrant workers returned home due to job losses, reducing remittance flows and further straining household incomes. These developments underline the relevance of examining Cambodia's agricultural response to COVID-19, particularly in the cassava sector, which plays a vital role in rural livelihoods. Understanding how cassava producers were affected by these shocks, and how they adapted, provides critical insights for building resilience in agriculture and informing more targeted, inclusive policy responses.

## 3.3 Methodology and Survey Framework

This paper analyzes cassava farming during the COVID-19 pandemic using descriptive statistics based on the household survey data collected in early 2022 by CDRI. The analysis explores patterns and trends in cassava production during the pandemic, focusing on production income, labor productivity, and gender dynamics. In this context, production income refers to the gross income derived from cassava sales prior to deducting production costs or other related expenditures for farming and households' consumption. Labor productivity is measured by the number of individuals participating in farming activities and the labor input per unit of cultivated land. Gender analysis examines the participation differences between male and female workers, both family members and hired labor, in cassava-farming operations.

### 3.3.1 Survey Location and Sample

The Cassava Household Phone Survey aimed to assess the impact of the COVID-19 pandemic on cassava production and farmers' socioeconomic conditions across Cambodia. Data collection was carried out with household heads or their spouses between mid-January and early February 2022. To ensure national representativeness, the study employed a stratified sampling approach. Five provinces, namely Tboung Khmum, Oddar Meanchey, Battambang, Banteay Meanchey, and Siem Reap, were selected, accounting for approximately 80% of the country's total cassava production area. Within each province, one district was randomly selected, and within those districts, a proportional approach was used to select 15 communes. Subsequently, where villages were randomly chosen within each commune. Collaboration with commune and village chiefs facilitated the creation of lists of households that produced cassava.[5] From this list, households were randomly selected for interviews. The final sample consisted of 768 households, with 384 households from Tboung Khmum, and the remaining 384 households distributed across the other four provinces. The larger sample size in Tboung Khmum reflects the province's significant role in cassava production.[6]

### 3.3.2 Survey Implementation and Challenges

The survey focused on two reference periods: the pre-pandemic year of 2019 and during the pandemic in 2021. These timeframes were chosen to ensure that respondents could recall relevant information accurately. However, conducting phone surveys during the pandemic presented several challenges. Studies have shown that phone surveys during lockdown periods often faced issues due to insufficient information, limited attention, and low response rates (Gourlay et al., 2021; Hirvonen et al., 2021). During this time, people prioritized their health, leading to missing information or inconsistent answers. Consequently, 60.8% of the original sample was excluded in the analysis, particularly households that did not report income, cost of cassava inputs, and/or labor participation for both 2019 and 2021.[7] Further data cleaning was necessary to address errors such as typographical mistakes or phone connection issues. To verify unclear information, follow-up calls were made to 50 respondents. However, 25% of respondents were either unreachable or unable

[5] The village chiefs sent the list of households in the form of image files. These lists were generally handwritten, some of which were difficult to read. The research team used Telegram chat or call to verify information with a village chief. The information contained in the list included the name of the household head or their spouse and phone number.

[6] Tboung Khmum alone accounts for 53.9% of the total cassava production area.

[7] In Cambodia, Eliste and Zorya (2015) illustrate that many farmers have low skills in management planning and financial management. This implies that many households have not adopted the practice of using notebooks to record their farming income and expenses. Consequently, farmers find it challenging to provide precise information regarding smaller farming-related details.

to provide clarifications on ambiguous answers. Notably, we found that households reporting the highest cassava gross income, production costs, and cultivated land sizes exhibited inaccuracies. To enhance reliability, we trimmed the top and bottom 5% of households based on cassava-related financial variables. As a result of these adjustments, the final dataset was reduced to 301 households, which meant that the sample was no longer nationally representative.

### *3.3.3 Income and Cost Calculation*

Utilizing CDRI's dataset, household gross income from cassava was calculated by multiplying the total quantity of cassava roots or chips sold (in tons) by the respective prices per kilogram (in riel). If a household sold both cassava roots and chips within the same year, the values were aggregated to determine total earnings. Similarly, the cost of cassava production were calculated by summing all relevant production expenses for each household, including input costs, labor expenses, transportation and storage costs, negotiation and communication fees, cassava chopping expenses, commission fees, plowing costs, land rental fees, and other associated expenditures.

## 3.4 Survey Results

This section presents the primary findings of the survey, showing changes in household cassava gross income, labor productivity, and gender participation in farming activities.

### *3.4.1 Cassava Gross Income and Production Cost*

Survey data indicate significant changes in households' gross income and cassava production expenditure between 2019 and 2021. As shown in Fig. 3.1, on average, cassava farmers across the five sampled provinces experienced a sharp decline in gross income, decreasing by 8.5% from $1,940.50 in 2019 to $1,774.50 in 2021. In contrast, production costs increased significantly by 9.3% from $1,073.30 to $1,173.30 (Fig. 3.1A). Among the samples, female-headed households had higher average gross income levels than the average male-headed households, with a gender income gap of 7.1% in 2019 and 6.1% in 2021 (Fig. 3.1B). Although female-headed households only account for 5.6% of the sample, they experienced a gross income decline of 9.3%, compared to an 8.5% decline for male-headed households. During the pandemic, there is a noticeable increase in expenditure in cassava production for both female- and male-led households. Female-headed households faced an increase

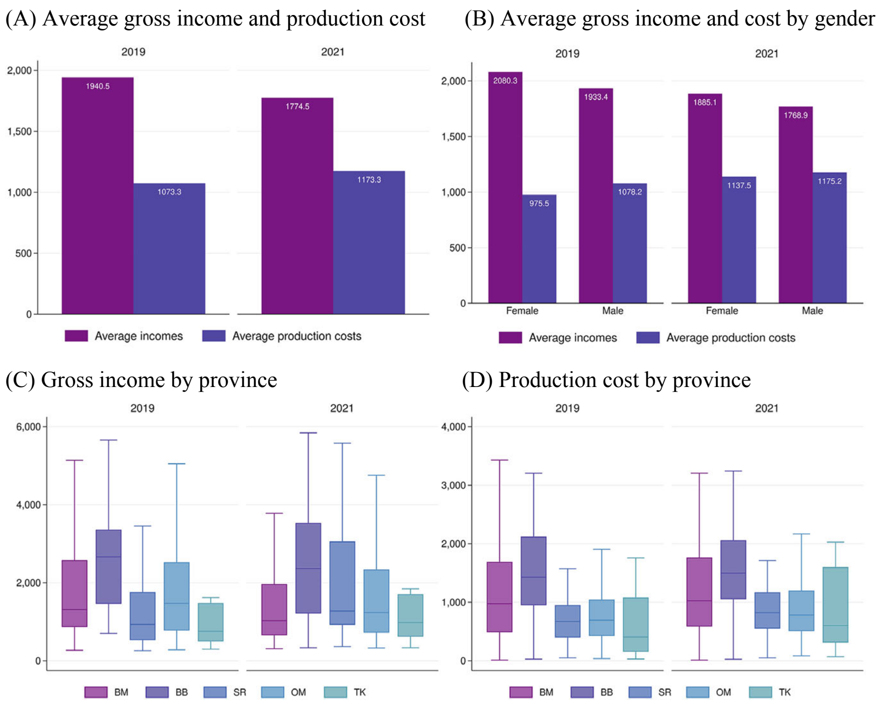


**Fig. 3.1** Households' gross income and cost of cassava production (*Note* n = 301 observations. The y-axis represents the value in US dollars based on the exchange rate between the riel and the dollar in 2019 and 2021. BM = Banteay Meanchey, BB is Battambang, SR = Siem Reap, OM = Oddar Meanchey, and TK = Tboung Khmum). *Source* CDRI Cassava Household Phone Survey 2022

of 16.6% (from $975.50 in 2019 to $1,137.50 in 2021) and 8.9% (from $1,078.20 in 2019 to $1,175.20 in 2021) for male-headed households.

At the province level, cassava gross income declined for farmers in Banteay Meanchey (23.3%), Oddar Meanchey (11.3%), and Battambang (7.4%). In contrast, farmers in Siem Reap and Tboung Khmum experienced substantial increases in gross incomes of 52.4% and 15%, respectively, indicating varied regional impacts of COVID-19 on farming. Production costs rose across all provinces, with the highest increase in Tboung Khmum, reported at 24.3%, followed by Siem Reap at 17.6%, Oddar Meanchey at 12.3%, and approximately 6% in Banteay Meanchey and Battambang.

Figure 3.2 illustrates how the pandemic impacts farmers' gross income and production costs based on cultivated land size. Households earning a gross income below $1,200 typically cultivated 2 hectares in 2019, and this value increased to an average of 2.3 hectares during 2021 (Fig. 3.2A). This suggests that in order to generate an income of approximately $1,200, farmers needed to increase their land

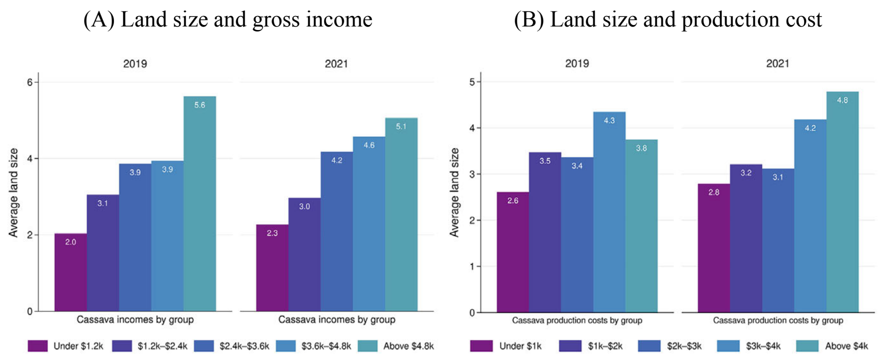


**Fig. 3.2** Average land size versus gross income and production cost (*Note* n = 301 observations. The y-axis represents the average cultivated land size in hectares). *Source* CDRI Cassava Household Phone Survey 2022

size by 0.3 hectares or 15%. Some farmers achieved this by renting land from neighboring farms, utilizing previously unused land, or purchasing new land. In 2019, farmers with an average land size of 3.9 hectares earned a gross income between $2,400 and $4,800. Notably, households within the $1,200 and $2,400, and above $4,800 brackets experienced a fluctuation in cultivation land, with some households maintaining the same planting size as in 2019, and others expanding their cropland. Land expansion trends varied within income groups, influenced by stable or rising cassava input prices in certain locations.

Rising production costs from COVID-19 also led to adjustments in cultivated land sizes (Fig. 3.2B). In 2019, a farmer could cultivate 3.5 hectares with a $1,000 investment. By 2021, this amount was insufficient, as increased production costs allowed cultivation of only 3.2 hectares. In such a situation, some farmers maintained similar land sizes but had to invest more capital in cassava production during the pandemic.

### 3.4.2 *Production Costs*

Farmers allocate their cassava production budgets based on several key components, encompassing expenses related to input use, labor, plowing, cassava chip cutting, transportation, negotiation and communication fees, and/or land rent. Figure 3.3 presents the average production costs incurred by households for each component. Typically, farmers allocate approximately $400 toward input use, covering chemical fertilizers, herbicides, stems, organic fertilizers, and insecticides or pesticides. The

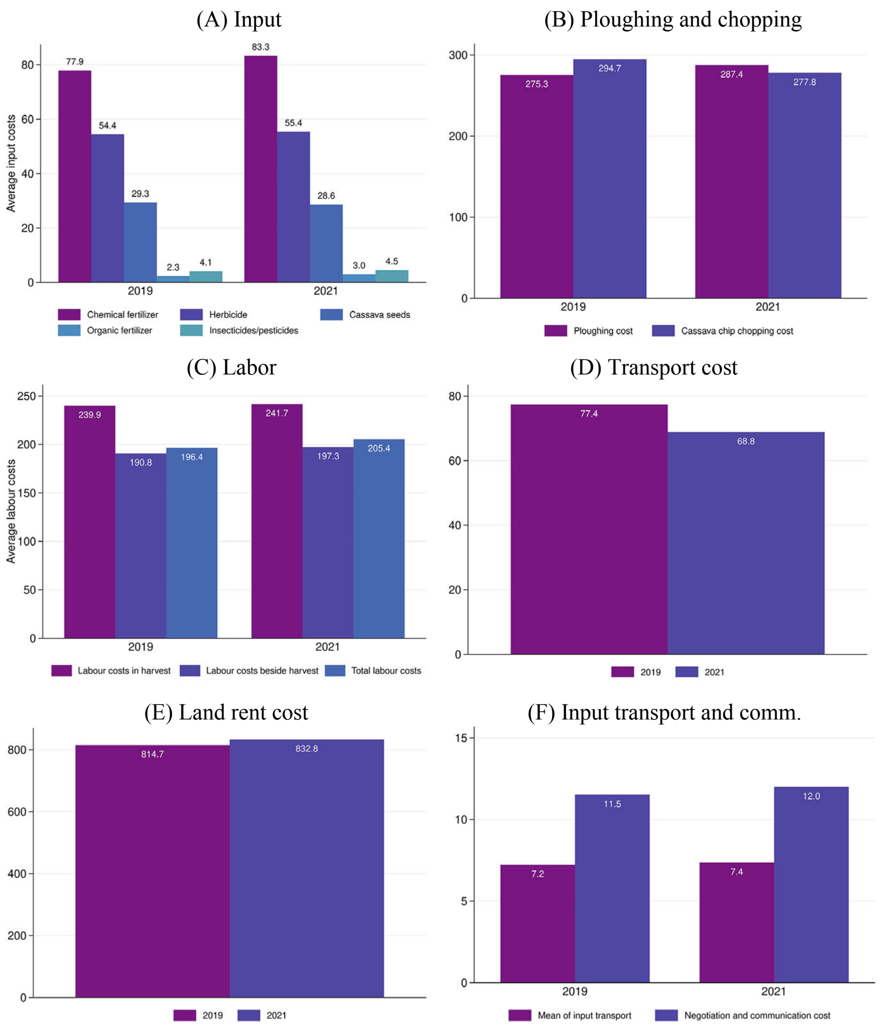


**Fig. 3.3** Production costs (*Note* n = 301 observations. The y-axis is the average input used in dollars based on the exchange rate between the riel and the dollar in 2019 and 2021). *Source* CDRI Cassava Household Phone Survey 2022

cost of these inputs varied based on cultivated land size and crop conditions. If households possess their stems from the preceding year, they can reduce input expenses.[8] In 2021, expenditures on organic fertilizer increased by 30.3%, chemical fertilizer

[8] From the survey, it was found that farmers mostly keep the seeds from the previous year but for the next production cycle, these seeds are not insufficient. Therefore, they need to purchase between 10 and 30% more for the next growing season.

by 7%, and insecticides or pesticides by 10.2% (Fig. 3.3A). However, expenditures on herbicides and cassava stems decreased by 1.8% and 2.5%, respectively.

The survey did not collect data on input prices or prices per unit,[9] making it challenging to accurately determine price fluctuations. However, according to Laborde (2023), the global fertilizer price index showed a major increase in the last quarter of 2021 and into 2022 and 2023. Consequently, Cambodian farmers were likely impacted by these global price fluctuations as the country predominately imports agricultural inputs.

Household expenditure on plowing and cassava chip processing constitutes a major cost in cassava production (Fig. 3.3B). While plowing costs increased by 4.4%, chopping costs declined by 5.7%. In addition, labor costs represent a significant portion of the total production cost (Fig. 3.3C). On average, labor expenses rose by 4.6%, from $196.40 in 2019 to $205.40 in 2021. Land rental costs also increased by 2.2%, from $814.70 to $832.80 in 2021.

### *3.4.3 Output Prices*

Farmers sell cassava in two forms: fresh cassava root and processed cassava chips. They sell these outputs to local traders, and the prices for these products vary across provinces. Households can sell cassava two or three times per seasonal production cycle in a year. Figure 3.4 shows price differences. For cassava root, the prices remained stable at the 2019 rate, averaging around 264 riels per kg for channel 1,228 riels/kg for channel 2, and 208 riels/kg for channel 3 (Fig. 3.4A). For cassava chips, the prices increased marginally between the pre-pandemic and pandemic periods. Selling cassava chips via channel 1, saw a price increase from 637 to 647 riels per kilogram. For channel 2, the price rose from 617 to 632 riels per kilogram and for channel 3, it increased from 601 to 631 riels per kilogram (Fig. 3.4B).

During the pandemic, farmers increased the sale of cassava roots by 5.8% (from 23.8 to 25.2 tons), while decreasing the sale of cassava chips by 9.2% (from 16.2 to 14.7 tons) (Fig. 3.5). The shift may be due to immediate cash needs, as fresh cassava can be sold more quickly than processed chips. When categorized by province, farmers in Battambang, Banteay Meanchey, and Siem Reap elevated the sale of fresh cassava, whereas those in Oddar Meanchey and Tboung Khmum experienced a decline (Fig. 3.5B). Similarly, mirroring the trend in cassava root sales, producers in Battambang and Siem Reap increased cassava chip sales, while those in Banteay Meanchey and Oddar Meanchey saw decreases.[10]

[9] It asked only the amount of money that households spend on inputs.

[10] The survey data lacks information on cassava chip sales from households in Tboung Khmum.

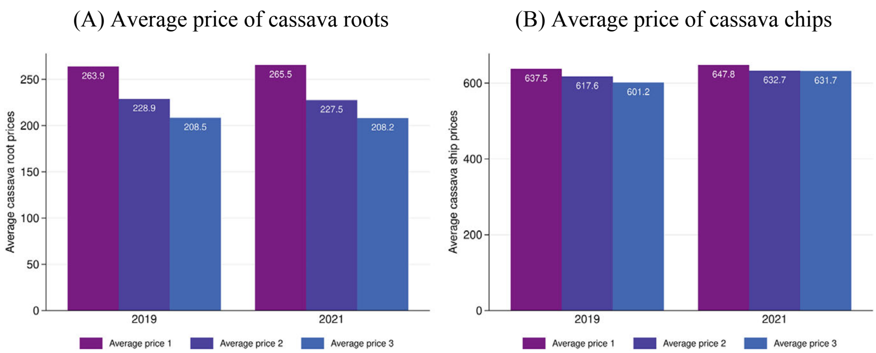


**Fig. 3.4** Output prices (*Note* n = 301 observations. The y-axis represents the average price of cassava output in dollars). *Source* CDRI Cassava Household Phone Survey 2022

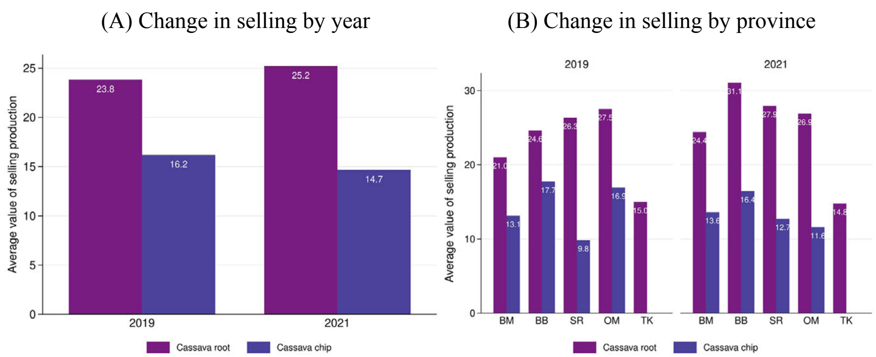


**Fig. 3.5** Average value of selling production (*Note* n = 301 observations. The y-axis represents the average value of selling cassava output in tons. BM = Banteay Meanchey, BB is Battambang, SR = Siem Reap, OM = Oddar Meanchey, and TK = Tboung Khmum). *Source* CDRI Cassava Household Phone Survey 2022

### *3.4.4 Changes in Productivity*

In Cambodia, household members play significant roles in farming activities. Figure 3.6 indicates the number of household members involved in cassava production and their average daily working hours. Across provinces, at least two household members engaged in cassava farming, and this pattern remained consistent even during lockdown periods (Fig. 3.6B). Migrant workers, particularly those engaged in garment and footwear, tourism, and the construction sector, as well as those who worked in Thailand (Huxtable, 2022),[11] returned to their homeland to shift

[11] According to the official data from the National Committee for Counter Trafficking, by the end of 2021, over 260,000 Cambodian migrant workers returned from abroad, mostly from Thailand, since the first outback of COVID-19 in March 2020 (Samean, 2021).

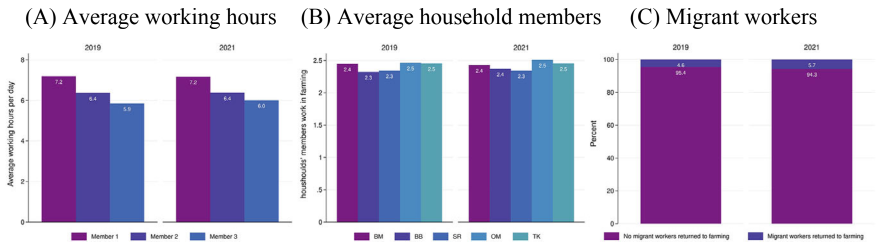


**Fig. 3.6** Household members in farming activities (*Note* n = 301. BM = Banteay Meanchey, BB is Battambang, SR = Siem Reap, OM = Oddar Meanchey, and TK = Tboung Khmum). *Source* CDRI Cassava Household Phone Survey 2022

to agricultural work. Our findings highlight an increase of 23.9% in migrant workers contributing to cassava-farming activities in 2021 (Fig. 3.6C).

Furthermore, households have increased their reliance on household members and have hired workers in cassava farming. The quantity of hired staff varied based on land size and demographic factors. Figure 3.7 shows that farmers cultivating land under 2 hectares, and between 2 and 4 hectares, typically employed 10 workers for cassava farming in 2019 (Fig. 3.7A,B). In 2021, this figure increased to 11 workers, and farmers extended the work period from 5 to 6 days for both cultivation and harvesting. Additionally, households with cultivated land exceeding 8 hectares also increased hired labor from an average of 11 workers before COVID-19 to 17 workers during the pandemic, averaging approximately 8 workdays in both the cultivation and harvesting periods.

In terms of gender distribution, female workers consistently constitute the majority of the labor force engaged in cassava farming. Based on our analysis, we find that hired female workers increased by 9.7% during the pandemic, while the number of hired male workers rose by around 10.8% (Fig. 3.7C). Average workdays during the cultivation and harvesting time for both genders increased, with females slightly higher at 7.5%, while for hired male workers, the increase was 7.2%. A detailed breakdown of the change in hired workers by province and gender is presented in the appendix.

## 3.5 Discussion

### *3.5.1 Interpretation of Findings*

This study highlights how Cambodia's cassava farmers were affected by pandemic-induced disruptions, revealing both short-term economic shocks and longer-term structural challenges in the sector. While some farmers reported increased labor availability due to the return of migrant workers from Thailand and other urban areas in Cambodia, overall productivity did not significantly improve. This suggests that

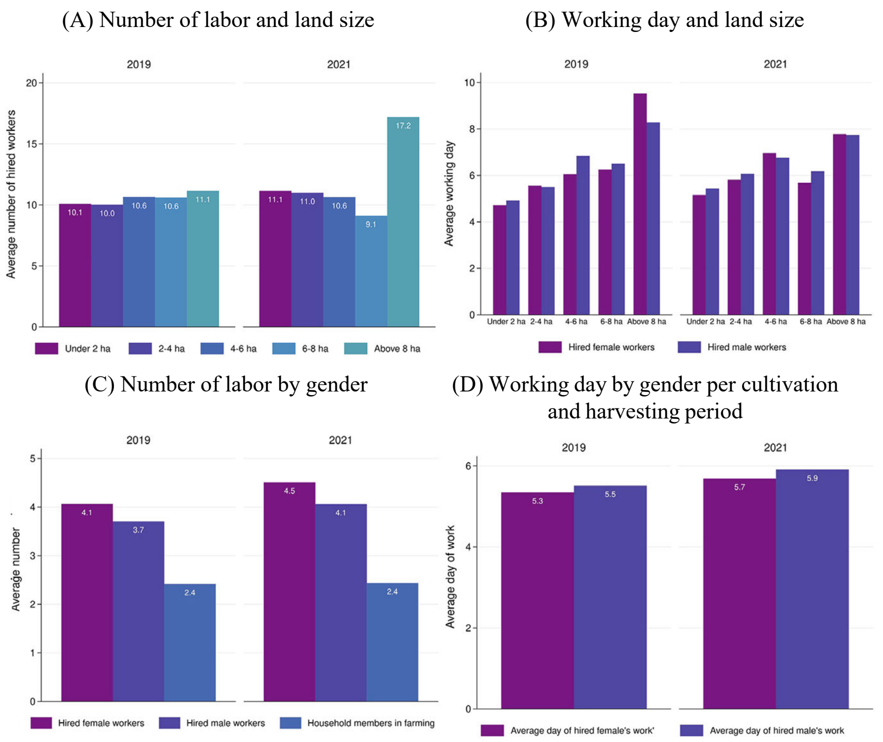


**Fig. 3.7** Change in productivity and labor participation per household (*Note* n = 301 observations). *Source* CDRI Cassava Household Phone Survey 2022

the influx of labor may have been low-skilled or temporary, rather than contributing to sustained efficiency improvements.

The findings also indicate that price volatility and export market dependence played a crucial role in farmer outcomes. Given that Cambodia exports a significant portion of its cassava, primarily to Thailand, Vietnam, and China, fluctuations in global demand and trade restrictions had an outsized impact on local farmers. This aligns with prior research showing that countries highly reliant on raw commodity exports often experience greater economic instability during global crises (Deaton & Miller, 1996; Eichengreen & Gupta, 2013; van der Ploeg & Poelhekke, 2009).

### *3.5.2 Comparisons to Regional and Global Trends*

Cambodia's experience mirrors that of neighboring Thailand and Vietnam, where cassava farmers also faced disruptions. However, in Thailand, government subsidies and price stabilization policies helped mitigate losses (Kongsil et al., 2024;

Sowcharoensuk, 2021). Similarly, Vietnam's greater focus on domestic processing and value addition allowed for better price resilience (PW Consulting Chemical & Energy Research Center 2025). In contrast, Cambodia's cassava sector remains heavily reliant on raw exports, leaving farmers vulnerable to price fluctuations and external trade policies (Keo et al., 2024).

Moreover, while global cassava prices rebounded in late 2021, Cambodian farmers may not have fully benefited due to limited storage capacity and an urgent need to sell during price downturns. This pattern has been observed in other commodity-dependent countries, where a lack of warehousing and access to financial credit forces farmers into distress sales at suboptimal prices.

### 3.5.3 Structural Challenges in the Cassava Sector

Beyond the impact of COVID-19, this study reveals several structural weaknesses within Cambodia's cassava sector. One key issue is the absence of robust processing industries, as the majority of cassava is exported in raw form, limiting the potential for value retention within the country. Investment in domestic processing facilities for products such as starch and bioethanol could significantly enhance the sector's resilience. Furthermore, while cassava is inherently drought-resistant, the sector remains vulnerable to the long-term impacts of shifting rainfall patterns and soil degradation. Climate adaptation strategies, including intercropping, the development of improved seed varieties, and soil conservation practices, are underexplored (Nith, 2021). In addition, Cambodia's heavy dependence on China and neighboring countries like Thailand and Vietnam as its primary export markets exposes the sector to significant risks. Diversifying export markets to regions such as South Korea, Japan, or the European Union could mitigate these risks and improve market stability, and importantly, producing cassava for local demand should be considered.

### 3.5.4 Policy Implications and Recommendations

The findings of this study highlight key policy priorities that are needed in order to strengthen Cambodia's cassava sector. While the Cambodian government took important steps during the COVID-19 pandemic, including providing low-interest agricultural loans through the Agricultural and Rural Development Bank, cassava received less targeted support compared to other crops. The National Cassava Policy 2020–2025 outlines a promising strategy, but implementation remains irregular and delayed, especially due to disruptions from the pandemic. Moving forward, several interventions are recommended:

- Increase Productivity through Training and Technology: Government ministries and development partners should support farmer training on climate-smart practices. Investments in research to identify high-yield, drought-resistant cassava varieties suited to local conditions are essential.
- Develop Processing and Value-Addition Capacity: Cambodia must reduce its reliance on raw cassava exports. Strengthening domestic processing industries for products like starch, ethanol, and animal feed can increase value retention and resilience against global price fluctuations.
- Improve Market Access and Infrastructure: Investments in transportation, storage, and digital market platforms are critical. Better logistics will allow farmers to time sales more effectively and reach new markets. Promotion of contract farming can also enhance supply chain coordination.
- Promote Inclusive and Gender-Responsive Development: Future policies should explicitly include women in cassava sector programs, from production to processing. Gender-responsive training and awareness campaigns can ensure equitable participation and broader social impact.
- Use Targeted Government Support During Crises: Financial safeguards such as emergency credit, tax relief, or price stabilization mechanisms should be introduced to protect smallholder farmers during periods of market disruption.

While the National Cassava Policy provides a comprehensive framework, accelerated implementation, stronger coordination, and inclusive, market-driven approaches are needed to realize its full potential. These policy actions can help Cambodia move from subsistence-level cassava production toward a high-value, resilient, and competitive agro-industrial sector.

## 3.6 Conclusion

Cassava stands as the second-largest agricultural product in Cambodia, playing a crucial role in rural income generation, particularly in regions near Thailand and Vietnam. However, the sector currently operates below its potential, largely reliant on raw exports and vulnerable to market shocks.

This study delved into the impact of COVID-19 on Cambodian cassava production. Our findings revealed that the pandemic led to a significant contraction in farmer income, despite stable cassava chip prices. In addition, we also find that farmers experienced increased production costs, limited processing capacity, and market dependency further constraining their resilience. These challenges underscore long-standing structural issues in the cassava sector, including low productivity, inadequate value addition, and exposure to international price volatility.

While Cambodia's National Cassava Policy 2020–2025 sets out a strong strategic vision, implementation gaps still remain. Accelerating investment in domestic processing, improving farmer's access to credit and training, and diversifying both the markets and the products is essential. As the country recovers from the pandemic and faces future global uncertainties, strengthening the cassava value chains offers a path for rural economic development and export competitiveness.

To strengthen Cambodia's cassava sector, future research should focus on several key areas: examining supply chain efficiencies, particularly how transport, storage, and logistics impact price volatility; exploring the potential of digital agriculture and market access, including whether mobile apps, blockchain, or digital marketplaces can enhance farmer incomes; and conducting comparative policy analysis to assess how Thailand and Vietnam's cassava policies differ and whether Cambodia can adopt similar strategies for growth and stability in the sector.

## Appendix

See Figs. 3.A1–3.A10.

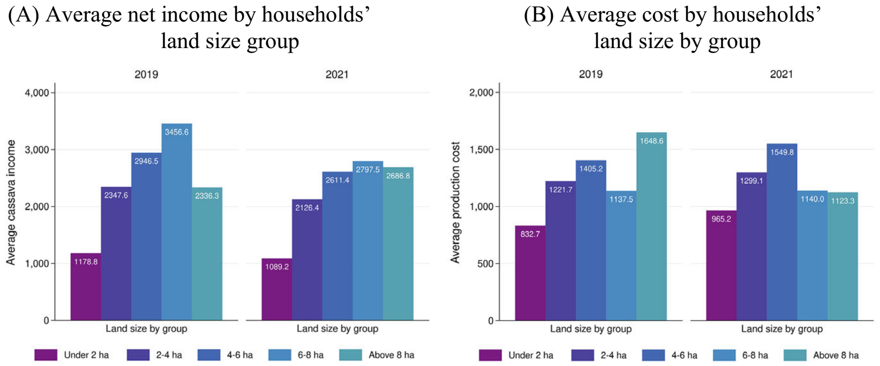


**Fig. 3.A1** Households' cassava net income and production cost by land size groups (*Note* The authors plot these figures by using the CDET's Cassava Household Phone Survey. We removed outliers in the 5 highest and lowest households' net income and expense on production. The sample used in these figures was 301 observations.)

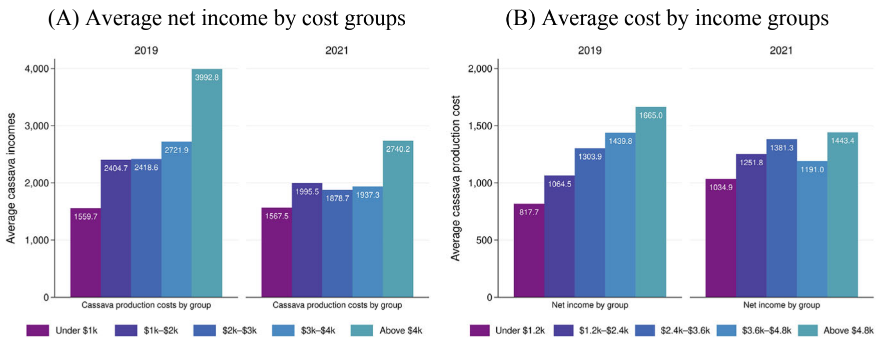


**Fig. 3.A2** Households' cassava net income and production cost (*Note* The authors plot these figures by using the CDET's Cassava Household Phone Survey. We removed outliers in the 5 highest and lowest households' net income and expense on production. The sample used in these figures was 301 observations.)

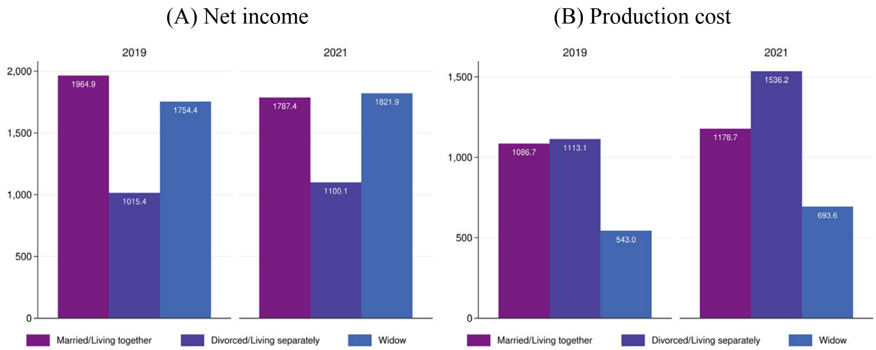


**Fig. 3.A3** Households' cassava net income and production cost by marriage status (*Note* The authors plot these figures by using the CDET's Cassava Household Phone Survey. We removed outliers in the 5 highest and lowest households' net income and expense on production. The sample used in these figures was 301 observations.)

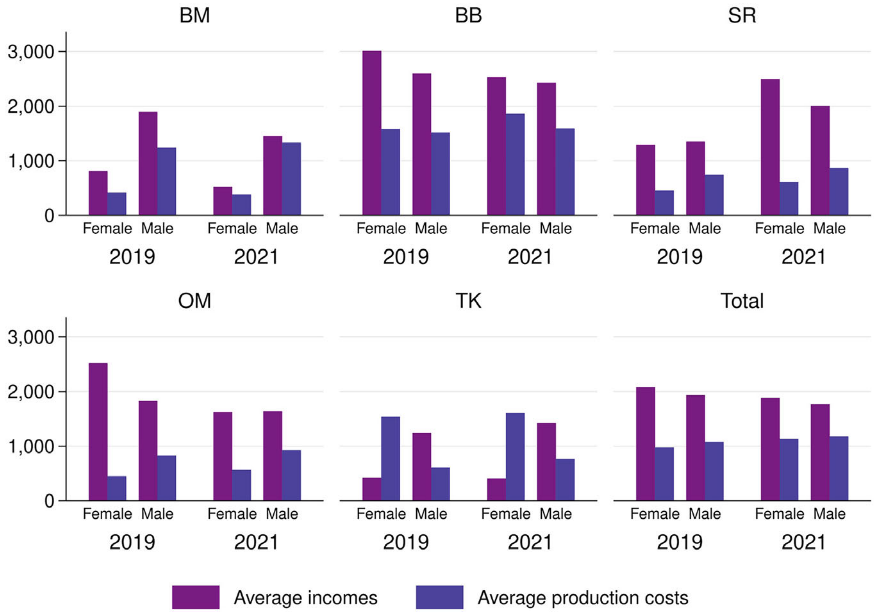


**Fig. 3.A4** Average households' cassava net income and production costs by gender of household head (*Note* The authors plot these figures by using the CDET's Cassava Household Phone Survey. We removed outliers in the 5 highest and lowest households' net income and expense on production. The sample used in these figures was 301 observations.)

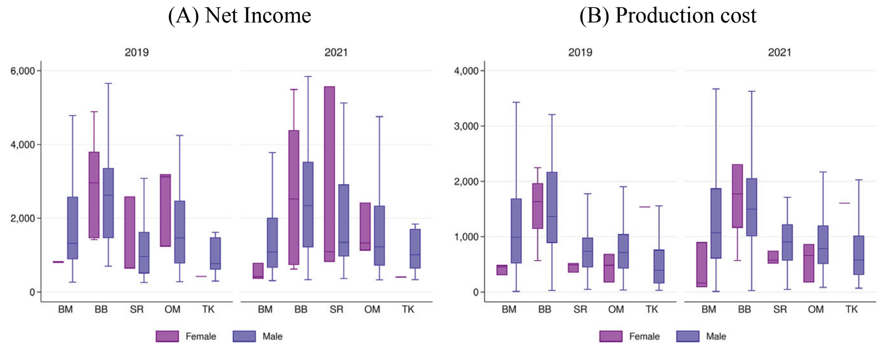


**Fig. 3.A5** Net income and production cost by gender of household head (*Note* The authors plot these figures by using the CDET's Cassava Household Phone Survey. We removed outliers in the 5 highest and lowest households' net income and expense on production. The sample used in these figures was 301 observations.)

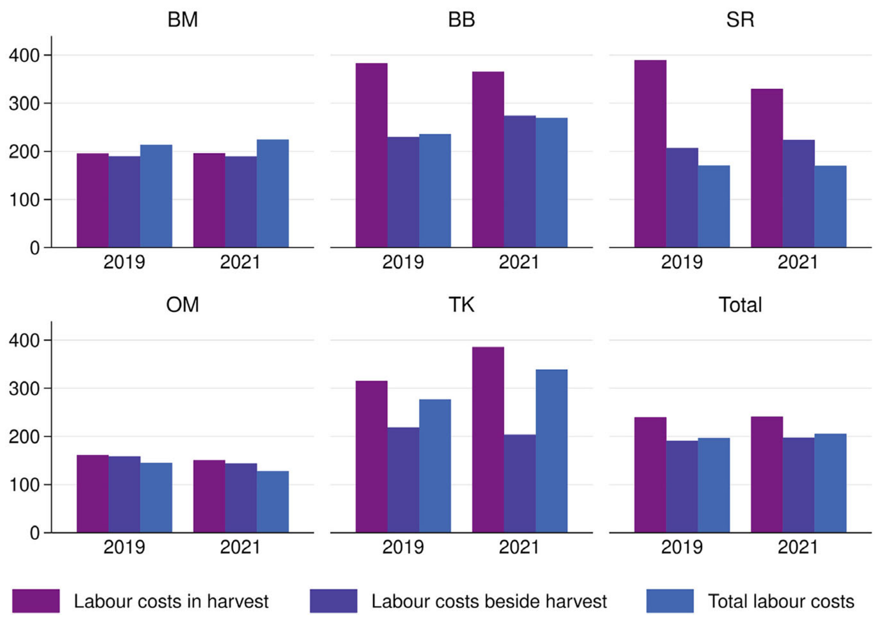


**Fig. 3.A6** Labor cost (*Note* The authors plot these figures by using the CDET's Cassava Household Phone Survey. The sample used in these figures was 301 observations.)

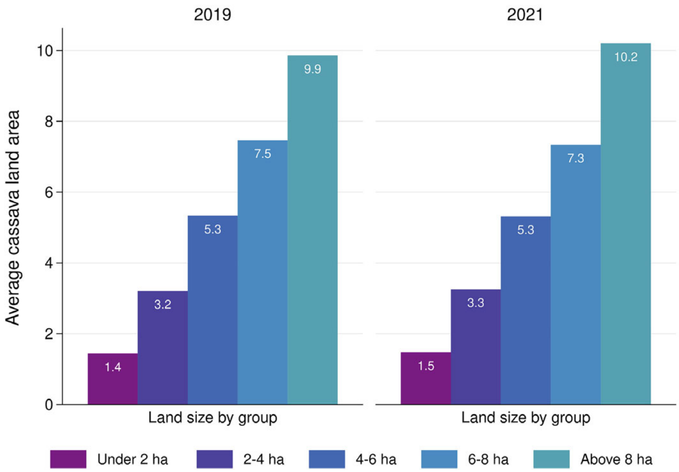


**Fig. 3.A7** Average land area by group (*Note* The authors plot these figures by using the CDET's Cassava Household Phone Survey. The sample used in these figures was 301 observations)

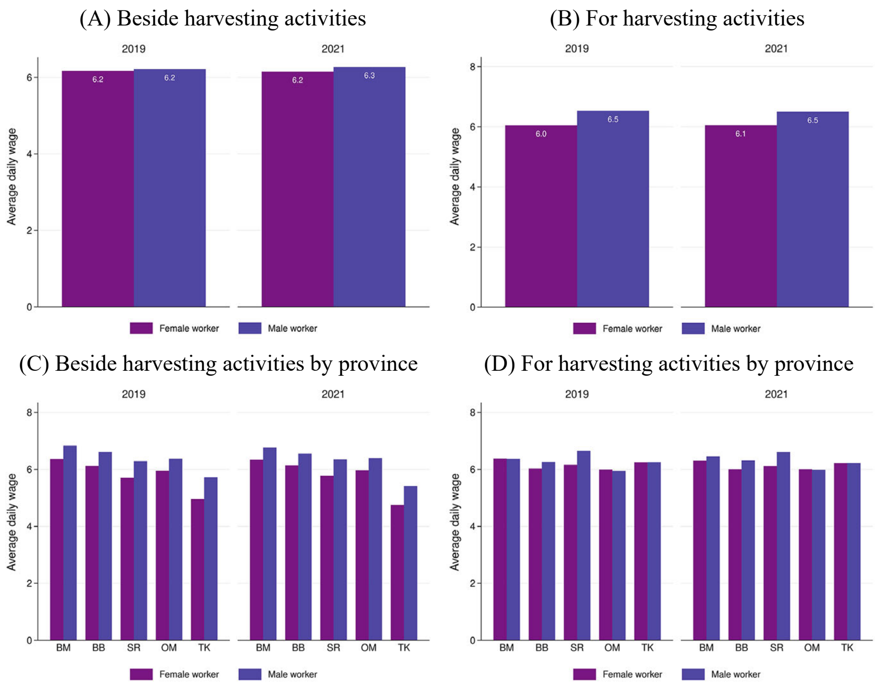


**Fig. 3.A8** Average daily wage for hired workers (*Note* The authors plot these figures by using the CDET's Cassava Household Phone Survey. The sample used in these figures was 301 observations.)

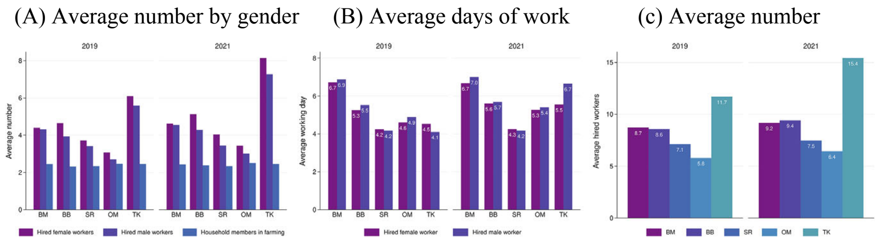


**Fig. 3.A9** Hired worked by provinces (*Note* The authors plot these figures by using the CDET's Cassava Household Phone Survey. The sample used in these figures was 301 observations.)


**Fig. 3.A10** Average interest rates per month (*Note* The authors plot these figures by using the CDET's Cassava Household Phone Survey. The sample used in these figures was 301 observations.)

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